1) An image appears too bright.

Grey level Transformation - Image Negative / Intensit
Reduction

=) A bright image has high intensity values that reduc
visibility

=) The most suitable grey level transformation is
image negative to decrease excessive brightness

=) It transforms pixel values using $s = L - 1 - r$, whe
L is the maximum

=) This redecus the brightness and improves
visiblity of image

=) Outcome: The image becomes balanced, and
hidden features are easier to observe

A dark image lacks detail.

Log Transformation.

=) A dark image contains low-intensity pixe
that hide important details.

=) Log transformation dark regions while pre
brighter area

=) The brightens dark regions while preservi[ng]
brighter area.
=) Outcome: Hidden details become visible, and
the overall image appears clearer

3) An image has poor constrast.

Histogram Equalization

=) Histogram equalization is used to improve
image contrast.
=) It redistributes the pixel intensity values over
the full gray-level range
=) This increases the difference between dark
and bright regions.
=) outcome: The image becomes sharper with
improved constrast and better visual
quality.

4) A low-contrast image requires uniform intensity
distribution.

Histogram Equalization

=) Histogram Equalization spreads pixel intensitie[s]
uniformly across the available range.

=> It calculates the cumulative histogram and maps old intensity values to new one.

=> This improves the overall constrant of the image

=> Outcome: Intensity values become more evenly distributed, making objects easier to identify.

5) An image shown uneven brightness across regions.

Adaptive Histogram Equalization (CLAHE)

=> Uneven brightness can be corrected using constrast limited Adaptive Histogram Equalization

=> CLAHE divides the image into small regions and enhances each region separately.

=> It also limits contrast amplificataion to avoid noise.

=> Outcome: Brightness becomes uniform across th image, and local details are enhanced without over-amplifying noise.